

EDS 222: Statistics for Environmental Data Science

"It is easy to lie with statistics. It is hard to tell the truth without statistics." - Andrejs Dunkels

 Instructor: Max Czapanskiy (they/them)

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 Office hours: Th 3-4 pm

 Office hours location: BH 3022A

 TA: Sloane Stephenson (she/her)

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 Office hours: Tu 2-3 pm

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 Lecture Time: MW 9:30-10:45

 Lecture Location: BH 1424

 Lab Time: Tu 12:30-1:50

 Lab Location: BH 3022A

I. Course Description

Statistics is the science of collecting, manipulating, and analyzing empirical data. In this class, we will learn the statistical fundamentals that will enable us to draw conclusions about the environment and its interaction with social and economic systems. We will cover fundamental statistical concepts and tools, and then apply and expand upon those tools to lay the foundation for temporal and spatial statistical methods that are particularly helpful in environmental data science. Welcome!

Course Website

<https://eds-222-stats-f26.github.io/>

All course content / information will be posted to this course website. Be sure to bookmark it somewhere you can easily find it!

II. Learning Goals

The goal of this course is to enable MEDS students to confidently and competently apply statistical tools to socio-environmental datasets. By the end of this course, students will be able to:

 **Summarize** data numerically and graphically

 **Interpret** random variables

 **Simulate** data to test assumptions

 **Describe models** using statistical notation

 **Draw DAGs** to clarify causal relationships

 **Quantify uncertainty** to perform inference

III. Course Format

EDS 222 uses a flipped classroom format. Before class, students will read assigned materials and watch recorded lectures. That will allow us to use time in class for case studies, applied activities, and drills.

Max will instruct during **lectures** on Monday and Wednesday mornings. These sessions will focus on building students' conceptual understanding of statistical methods and techniques. Please come prepared to participate in activities and discussions. *It is essential that students complete the readings and recorded lectures in advance.* Bring paper, pencil, your laptop, and (most importantly) your questions.

Sloane will instruct during **labs** on Tuesday afternoons. These sessions will dive deeper into the concepts and case studies from the lectures. Expect hands-on activities, problem sets, and working in R.

IV. Course Content

Below is a table of the topics and deliverables in EDS 222. Pre-recorded lectures, slides, and activities will be posted to the course website.

Readings come from the following open educational resources, available online. **Complete readings and pre-recorded lectures before class.**

[[ROS](#)] Gelman, A., Hill, J., & Vehtari, A. (2021). *Regression and Other Stories*. Cambridge University Press.

[[IMS](#)] Çetinkaya-Rundel, M., & Hardin, J. (2024). *Introduction to Modern Statistics* (2e ed.). OpenIntro.

[[CIR](#)] Cunningham, S. (2026). *Causal Inference: the Remix*. Yale Press.

 Week	 Topic	 Deliverables	 Readings	
1	Exploring data		Monday No reading	Wednesday ROS 1-1.4, 2-2.2 IMS 1-2
2	Foundations of inference	HW1 / Exploratory analysis	Monday IMS 11-13	Wednesday IMS 16-17, 19-21
3	Linear regression		Monday IMS 7	Wednesday No reading
4	Multiple regression	HW1 revisions	Monday IMS 8	Wednesday CIR 3
5	Linear model inference	HW2 / Blog post proposal	Monday IMS 24	Wednesday IMS 25
6	t-test, ANOVA, F-test		Monday No reading	Wednesday No reading
7	Logistic regression	HW2 revisions	Monday IMS 9	Wednesday IMS 26
8	Generalized linear models	Midterm	No reading	
9	Spatio-temporal methods	HW3 / Blog post first draft	No reading	
10	Statistics in action!	Midterm corrections	No reading	
Finals	NA	HW3 revisions / Blog post final draft		

V. Grading

Grades for EDS 222 are based on four components.

Participation (10%). Students will be assessed based on their active participation in classroom activities, weekly reflections, and contributions to the pre-class “parking lot”. This is a living Google Doc where students can post questions and reflections about the readings and recorded lectures.

Homework (30%). In three homework assignments, students will implement the statistical tools from lecture and lab in R. Homeworks are due at 11:59pm on the Sundays following weeks 2, 5, and 9. Homeworks will be distributed through GitHub Classroom; the links will be posted on the course website. **Revisions are due one week after grades are posted, and can earn up to 50% of missed points.** For example, Angela submits HW1 at the end of week 2. She receives feedback in week 3 and gets an 80% on the assignment. During week 4, she revises her assignment and fixes all of the issues. Her revisions earn her half of the missed points, increasing her grade to 90%.

Midterm (30%). There will be one midterm held in class on Wednesday, November 18th. The format of the midterm is ten shorter questions and two longer, multi-part questions. Students are permitted to bring one 8.5x11" piece of paper with notes (front and back). **As with the homework assignments, students will have an opportunity to make corrections, due one week after grades are posted, and earn up to 50% of the missed points.**

Blog post (30%). The final project is a blog post that applies concepts and tools from class to the investigation of a question of the students' choosing. There are multiple deliverables over the quarter to provide scaffolding for the final project. **Only the final draft of the blog post is graded**, but the intermediate deliverables provide valuable opportunities to assess project feasibility, manage time, and organize thoughts.

VI. Attendance Policy

In-person attendance to classes and discussion sections is crucial as it allows you to engage in hands-on activities while minimizing distractions, ask questions in real-time, and form connections with your peers and instructor. These experiences are designed to help you grow professionally! **Regular attendance also mirrors professional expectations** in many workplaces and is needed to grow relationships that can become valuable professional references.

Similar to missing a day at work, if you miss a class you are expected to:

- Be proactive: Notify the instructor before it happens or within a day and provide a brief explanation. Documentation is not required for missing a class, but honesty and professionalism is expected in all communication.
- Catch up: Review course material on the website with a peer to cover any missed material. Identify where you have gaps or muddy points of understanding, and follow up with your instructors.
- Stay home when you are sick! If you are feeling unwell, prioritize your health. Stay home and rest or work remotely.

Attendance does not count towards your grade, but it will be tracked and absences without communication will be addressed.

UCSB courses are taught in person, so absences for two or more weeks may require a [Leave of Absence](#). If you have to miss multiple classes, contact the instructor, the Director of Student Services ([Kristine Faloon](#)), and the MEDS Program Coordinator ([Jamie Montgomery](#)) to discuss possible next steps on how to support your well-being and progress within the program.

There will be no option for remote attendance, except for the week of Thanksgiving break. Other accommodations will need to be carefully considered for equity. Eligible students should work with the Disabled Students Program (DSP) to determine appropriate accommodations.

VII. AI Policy

The use of Large Language Models (LLMs) and other generative AI tools is becoming more widespread in environmental data science. Research on the effects of generative AI on learning indicates that they are best avoided at this stage in your education. ***Generative AI shortcuts your development by offloading the mental struggle required to build understanding.*** While it may be helpful in generating short-term solutions, studies highlight that using AI to generate answers during early skill-building can hinder long-term skill development and independent problem-solving ([Bastani et al., PNAS, 2025](#)). Practicing involves making mistakes, and exploring solutions to gain understanding. We will move step-by-step, making sure you gain confidence, independence, and mastery through your own problem-solving and collaboration.

AI-Generated Code Policy

All code submitted in this course should be written by the student. If a student submits code that appears to be generated by an LLM, I will ask them to meet outside of class and describe the purpose and function of the code verbally. If the student cannot satisfactorily explain the code, they will not receive credit for the associated assignment. If it happens a second time, that will constitute a violation of the academic [code of conduct](#).

Your instructional team is here to support you every step of the way. By tackling challenges directly, practicing, and asking questions, you'll gain the confidence and skills to succeed both in this course and in your future work!

VIII. Campus Resources

There are many on-campus resources that support graduate students in navigating challenges, building community, and staying well during the quarter. Your instructors are also available to discuss individual needs and help connect you with a campus resource that may be best suited to your situation.

- 🍎 **Food Insecurity:** UCSB has several food resources available. Explore options on their [website](#), including the [CalFresh Program](#) (aka SNAP benefits) and the [Associated Students food bank](#).
- 🏠 **Housing Needs:** As a student of UCSB and a member of the community, you should have access to stable housing. Browse [UCSB's resources](#), including [emergency transitional housing](#) support.
- 🧠 **Mental Health:** UCSB's [Counseling and Psychological Services \(CAPS\)](#) provides confidential mental health services and crisis support. Their 24/7 phone number is (805) 893-4411.
- 💰 **Financial Crisis:** If you are experiencing a financial crisis, email financialcrisis@sa.ucsb.edu. Within 1-2 days, a financial advisor will reply. UCSB also has [short-term emergency loan](#) programs.
- 🦋 **Undocumented Students:** [Undocumented Student Services \(USS\)](#) offers workshops, helps students find scholarships and financial support, and works to provide a campus community.
- ❤️ **Interpersonal Violence:** The [Campus Advocacy, Resources, & Education \(CARE\)](#) office has 24/7 confidential support and advocacy for sexual assault, dating and domestic violence, and stalking.
- 🌈 **Gender and Sexuality:** The [Resource Center for Sexual and Gender Diversity \(RCSGD\)](#) office and the [Queer & Trans Graduate Student Union \(QTGSU\)](#) hosts events & resources for LGBTQ+ students.

- 🏥 **Medical Health:** All UCSB students, including master's students, have full access to the [campus health center](#), including a free after-hours nurse line at (877) 351-3457 and social work services.
- ❌ **Interpersonal Conflict:** The [Office of the Ombuds](#) is a safe and confidential place to discuss workplace issues, interpersonal conflict, academic concerns, and other campus-related problems.
- 🧠 **Addictive Behaviors and Recovery:** The [Addictive Behaviors & Recovery \(ABR\)](#) promotes safe, healthy choices around substance use and offers free, confidential counseling to all students to reduce high-risk behavior.
- 🌍 **Cultural Community:** The UCSB [Multicultural Center \(MCC\)](#) is a community space that provides critical dialogue and upholds a safer, welcoming space that validates marginalized identities.
- ❤️ **Black Community:** The [Black Graduate Student Association \(BGSA\)](#) is a UCSB organization that provides academic support and social resources support that meet the needs of Black students.
- 🍷 **Indigenous Community:** The [American Indian and Indigenous Cultural Resource Center \(AIICRC\)](#) provides professional development opportunities, scholarships, and community.
- 💙 **Asian & Pacific Islander Community:** The [Asian Pacific Islander Graduate Student Alliance \(APIGSA\)](#) strives to support graduate students of Asian and Pacific Islander panethnicity.
- 💚 **Hispanic & Latinx Community:** The [Society for Advancement of Chicanx and Native Americans in Science \(SACNAS\)](#) includes a graduate chapter for the UCSB community that hosts events.
- 👩 **Women in STEM:** UCSB has a graduate chapter of [Women and Science in Engineering \(WISE\)](#). They host events, career panels, and regular K-12 outreach events with the community.
- 🎓 **Graduate Advocacy:** The [Graduate Student Association \(GSA\)](#) is the elected representative government for students and includes funding for childcare support, emergency grants, & travel.

IX. Instructional Team

Instructor: Max Czapanskiy is an Assistant Teaching Professor in Marine Data Science, jointly appointed to UCSB's Bren School of Environmental Science & Management and the College of Creative Studies. A former software engineer turned marine ecologist, they apply the tools and problem-solving strategies from computer science to questions about the ecology and conservation of large marine vertebrates. They earned degrees in Computer Science, Geographic Information Systems, and Marine Biology while working for corporations (Microsoft), federal agencies (USGS and NOAA), and non-governmental organizations (Point Blue Conservation Science). When not at a keyboard or a whiteboard, they can most often be found in a kitchen baking bagels or trying a new recipe (preferably involving dried chiles). At UCSB, they teach MEDS courses and advise Capstone Projects in the Bren School, and teach undergraduate Marine Science majors in the College of Creative Studies.

Teaching Assistant: Sloane Stephenson (she/her)

Sloane Stephenson is a PhD candidate at Bren, where she studies wildfires and post-fire ecosystem recovery in California's Central Coast. Using the ecohydrological model RHESSys, her dissertation investigates how climate regimes and plant ecophysiological traits influence vegetation recovery, groundwater recharge, and fire regimes in Santa Barbara County. She also studies how different climate model inputs influence projections of fire size, fire intensity, and post-fire recovery trajectories, with the goal of informing regional wildfire resilience strategies in the face of climate change. Before coming to Bren, Sloane worked as a data scientist in the insurance industry, and she is passionate about showing students that quantitative and analytical backgrounds can serve as a powerful entry point into environmental research and problem solving. When she isn't in the classroom (or squinting at her laptop), she is most likely spending quality time with her cat Juneau or enjoying the outdoors!